By Interacting Object (Box, Sphere, Cylinder) — Motions: Kinematic (Horizontal, Vertical, Circle, Hold) & Dynamic (Drop, Slide, Throw, Tumble)

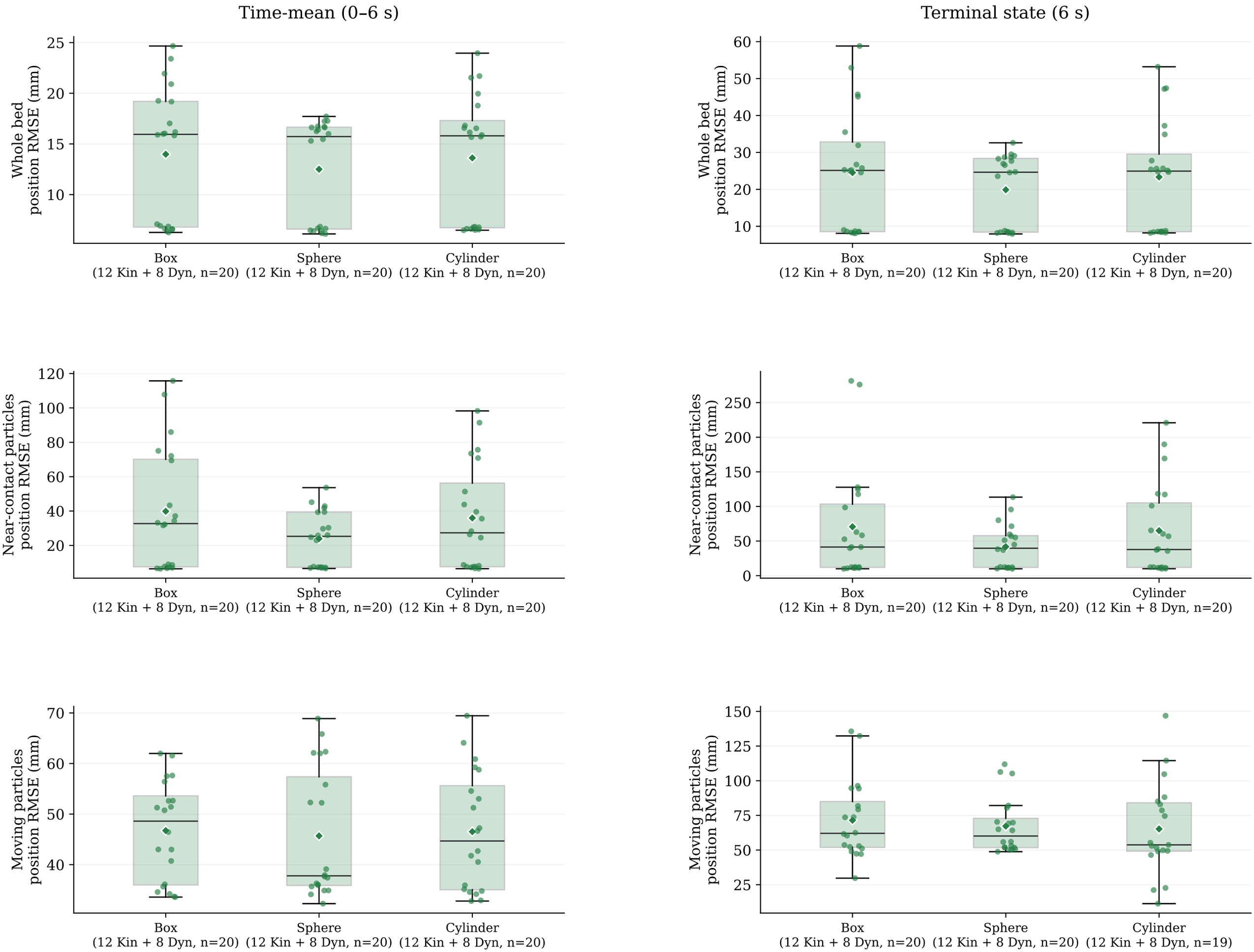

Figure 1a: Frozen object-model checkpoint 10,800. Elementary objects (Box, Sphere, Cylinder), each aggregated across 20 trajectories: 12 Kinematic (Horizontal, Vertical, Circular, Hold) and 8 Dynamic (Drop, Slide, Throw, Tumble). Rows show particle-position RMSE for the entire bed, near-contact particles (≤0.05 m), and moving particles (>0.01 m/s). Left column: time-mean over 0–6 s; right column: terminal state at 6 s.